WORKED SOLUTION

STEP I 2009, Question 11

Mechanics • Collisions and Restitution

STEP I 2009, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/09-s1-q11

The key idea

Every one-dimensional collision is governed by two equations: conservation of momentum and Newton's law of restitution. The trick for the energy is to write each particle's loss as a difference of two squares, $\frac{1}{2}m(u-p)(u+p)$, because momentum conservation ties the factors $u-p$ and $q-v$ together. Once a common factor $u-p$ has been pulled out, the remaining bracket is exactly what the restitution law describes.

Solution

We model the particles as moving along a line on a smooth table, with velocities measured in the direction of $\mathbf{i}$. The particle of mass m is behind and catches up, since $u > v$. With no horizontal external forces, momentum is conserved:

$$mu + Mv = mp + Mq, \qquad \text{so} \qquad M(q-v) = m(u-p). \tag{1}$$

Newton's law of restitution says the speed of separation is e times the speed of approach:

$$q - p = e(u-v). \tag{2}$$



before (velocities along $\mathbf{i}$), $u > v$

Part (i)

The loss of kinetic energy is

$$\Delta = \tfrac{1}{2}m(u^2-p^2) + \tfrac{1}{2}M(v^2-q^2) = \tfrac{1}{2}m(u-p)(u+p) - \tfrac{1}{2}M(q-v)(q+v).$$

By (1) we may replace $M(q-v)$ by $m(u-p)$, so a factor $\frac{1}{2}m(u-p)$ comes out:

$$\begin{aligned}\Delta &= \tfrac{1}{2}m(u-p)[(u+p)-(q+v)]\\ &= \tfrac{1}{2}m(u-p)[(u-v)-(q-p)]\\ &= \tfrac{1}{2}m(u-p)[(u-v)-e(u-v)] \qquad \text{by (2)}\\ &= \tfrac{1}{2}m(u-p)(u-v)(1-e),\end{aligned}$$

as required.

Ask yourself How does a formula for the energy loss tell us anything about p?

A collision can't create kinetic energy, so $\Delta \geq 0$. In the product $\frac{1}{2}m(u-p)(u-v)(1-e)$ we have $m>0$, $u-v>0$ (given) and $1-e>0$ (since $e<1$). So the sign of Δ is the sign of $u-p$, and $\Delta \geq 0$ forces $u \geq p$. The particle of mass m can't speed up in the direction of $\mathbf{i}$, which makes sense since the impulse on it from M acts backwards.

If you'd rather not rely on $\Delta \geq 0$ as a physical fact, you can solve (1) and (2) directly:

$$p = \frac{mu+Mv-Me(u-v)}{m+M}, \qquad \text{so} \qquad u-p = \frac{M(1+e)(u-v)}{m+M} > 0$$

for $e \geq 0$. That confirms $u \geq p$, and hence $\Delta \geq 0$.

Part (ii)

Each particle loses the same non-zero amount of energy, so

$$\tfrac{1}{2}m(u-p)(u+p) = \tfrac{1}{2}M(v-q)(v+q) \neq 0.$$

By (1), $M(v-q) = -m(u-p)$, so the right-hand side is $-\frac{1}{2}m(u-p)(v+q)$ and

$$\tfrac{1}{2}m(u-p)(u+p) = -\tfrac{1}{2}m(u-p)(v+q).$$

The left-hand side is non-zero, so $m(u-p) \neq 0$ and we may divide by $\frac{1}{2}m(u-p)$:

$$u+p = -(v+q), \qquad \text{that is,} \qquad u+v+p+q = 0. \tag{3}$$

Now (2) and (3) give the sum and difference of p and q:

$$p+q = -(u+v), \qquad q-p = e(u-v),$$

so

$$p = -\tfrac{1}{2}[(u+v)+e(u-v)], \qquad q = -\tfrac{1}{2}[(u+v)-e(u-v)].$$

We haven't used momentum conservation in its full form yet, and that is what fixes e. Substituting into $mp+Mq = mu+Mv$ and multiplying by 2:

$$\begin{aligned}-(m+M)(u+v)+(M-m)e(u-v) &= 2mu+2Mv,\\ (M-m)e(u-v) &= (m+M)(u+v)+2mu+2Mv\\ &= (M+3m)u+(3M+m)v.\end{aligned}$$

Since $u - v > 0$, and $M - m \neq 0$ when $m \neq M$, we can divide to get

$$e = \frac{(M+3m)u + (3M+m)v}{(M-m)(u-v)},$$

as required.

Checks

- Take $m = 1$, $M = 3$, $u = 1$, $v = -\frac{1}{2}$. The formula gives $e = \frac{6-5}{2\times\frac{3}{2}} = \frac{1}{3}$. Then $p = -\frac{1}{2}$ and $q = 0$, and momentum checks out: $1 - \frac{3}{2} = -\frac{1}{2} + 0$. Each particle loses $\frac{3}{8}$ of energy, and $u + v + p + q = 1 - \frac{1}{2} - \frac{1}{2} + 0 = 0$.
- For $e = 1$ the formula in (i) gives zero loss, as it should for a perfectly elastic collision.
- The computer tested (i) with exact fractions on 2000 random collisions, and (ii) on 455 random cases with $0 \leq e < 1$.

Common slips

- Getting the restitution law back to front: it's $q - p = e(u - v)$, separation over approach, with both measured as positive quantities here.
- Dividing by $u - p$ without saying why it's non-zero. Here that's exactly what "non-zero amount" is there for.

Beyond STEP

The factorisation in (i) is a neat consequence of working in the centre-of-mass frame. In that frame the total momentum is zero, the kinetic energy is $\frac{1}{2}\mu(\text{relative speed})^2$ with reduced mass $\mu = \frac{mM}{m+M}$, and the loss is $\frac{1}{2}\mu(1-e^2)(u-v)^2$. Reduced mass and the centre-of-mass frame turn up again in two-body orbits and in particle scattering. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops these ideas carefully.

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